

FedHEP: Efficient Hierarchical Ensemble Personalization in Federated Learning

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Abstract

Federated Learning (FL) on resource-constrained edge devices faces a fundamental trilemma: mitigating client drift under non-IID data, enabling personalization without dual-model memory overhead, and defending against Byzantine poisoning without penalizing honest specialized clients. We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), a single-backbone framework coordinating three representation tiers—global consensus, collaborative peer clusters, and private local heads. First, Active-Class Logit Masking (**ACLM**) with skew-calibrated loss weighting eliminates gradient interference from unobserved categories without duplicating feature extractors. Second, low-dimensional Random Projections (**RP**) enable privacy-preserving cluster discovery among peers. Third, Subspace-Constrained Cosine Filtering (**SCCF**) coupled with Temporal Trust Tracking (**TTT**) enhances Byzantine resilience under moderate corruption by leveraging active-class alignment to reduce false-positive rejection of specialized clients. Experiments on CIFAR-100 with ResNet-9 across five Dirichlet non-IID regimes, multi-attack Byzantine benchmarks, and 50-client scaling show that FedHEP improves personalization accuracy and worst-decile tail fairness over competitive baselines. Physical execution profiling on MobileNetV3 confirms single-model memory footprint and latency efficiency on resource-constrained edge platforms.

Keywords

Federated Learning, Multi-Scale Personalization, Byzantine Fault Tolerance, Statistical Heterogeneity, Tail Fairness

ACM Reference Format:

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1 Introduction

Federated Learning (FL) has emerged as the preeminent paradigm for privacy-preserving distributed machine learning, enabling massively distributed edge clients to collaboratively train neural models without centralizing raw telemetry [15, 21]. In realistic deployments on resource-constrained embedded nodes (e.g., IoT sensors, autonomous micro-agents, mobile edge devices), multiagent collaborative learning encounters a fundamental *trilemma* spanning statistical heterogeneity, edge hardware boundaries, and adversarial threats.

First, real-world edge data distributions across participating nodes are fundamentally non-independent and identically distributed (**non-IID**), frequently exhibiting severe label skew [13]. Under such distribution shifts, conventional consensus aggregation schemes such as FedAvg [21] and FedProx [19] suffer from severe *client drift*, where gradient updates pull the shared parameters toward orthogonal local optima, catastrophic forgetting ensues, and tail performance across specialized nodes collapses [20].

Second, while personalized FL has been proposed to overcome client drift, existing formulations incur prohibitive on-device overheads. Dual-model architectures (such as Ditto [18]) maintain concurrent global and local network instances, virtually doubling parameter footprint, forward-backward graph allocations, and peak VRAM usage on hardware-constrained edge nodes. Conversely, head-decoupling frameworks like FedRep [5] bifurcate representation learning into disjoint global backbone and local head phases, yet leave local heads isolated from peer collaboration and prone to overfitting on sparse local support.

Third, the threat of Byzantine failures and adversarial model poisoning presents a critical failure mode in distributed multiagent systems [2, 8, 24]. Standard Byzantine-robust aggregation rules (e.g., Multi-Krum [2], coordinate-wise median, and trimmed mean) compute global geometric distances in the unconstrained parameter space. Under extreme label skew, honest updates from highly specialized clients exhibit legitimate angular deviations that standard Euclidean filters conflate with malicious poisoning, erroneously penalizing honest nodes while admitting stealthy in-distribution adversaries.

To resolve this trilemma, we present **FedHEP** (*Federated Hierarchical Ensemble Personalization*), an efficient framework designed for resource-constrained multiagent edge topologies. Specifically, our key technical contributions are:

- **Single-Backbone 3-Tier Representation Hierarchy:** We design an architecture where a single shared feature extractor concurrently feeds three linear classification tiers: a global consensus Root head (W_r), a collaborative cluster Parent head (W_p), and a private Local head (W_l), completely eliminating parameter and memory duplication.
- **Interference Mitigation via ACLM & Skew-Calibrated Weighting:** We introduce Active-Class Logit Masking (**ACLM**) and an anchored dynamic binomial loss weighting scheme, eliminating negative backpropagation drag from unobserved labels onto the shared feature extractor.
- **Privacy-Preserving Dynamic Peer Clustering:** Clients dynamically identify collaborative clusters via low-dimensional Random Projections (**RP**, $m = 256$) bounded by the Johnson-Lindenstrauss lemma [14], clustering clients by functional similarity without exposing raw local gradients.



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- **Layered Subspace Byzantine Defense:** We develop a multi-tier defense combining Subspace-Constrained Cosine Filtering (SCCF) restricted strictly to each client’s observed class support, backbone update norm-bounding, and multi-round Temporal Trust Tracking (TTT).
- **Rigorous Empirical Validation:** To realistically capture edge client data scarcity rather than datacenter-scale silos, we evaluate on edge-budgeted partitions of CIFAR-100 (10,000 train / 3,000 test) and FEMNIST (15,000 train / 3,000 test) across five Dirichlet non-IID regimes ($\alpha \in [\infty, 0.05]$) and 50-client scaling. We demonstrate superior personalization accuracy and worst-decile tail fairness against 6 state-of-the-art (SOTA) baselines, resilient defense across 4 Byzantine poisoning vectors, and minimal on-device footprint via physical execution profiling on MobileNetV3.

2 Related Work

To combat client drift under non-IID statistical heterogeneity, personalized federated learning departs from monolithic global consensus. Consensus-regularization schemes like FedProx [19] and SCAFFOLD [16] stabilize optimization via proximal terms or control variates, yet still restrict clients to a single consensus hypothesis. Meta-learning frameworks such as Per-FedAvg [7] train an initial meta-model for fast local adaptation, but incur computationally heavy second-order Hessian calculations. Interpolation and dual-model architectures (e.g., APFL [6] and Ditto [18]) decouple global and personalized objectives by maintaining two distinct concurrent networks per client, effectively doubling on-device memory and gradient overhead. Head-decoupling formulations (e.g., FedPer [1], FedRep [5], FedBABU [22], and FedALA [25]) isolate classifier heads from a shared backbone, yet leave localized heads entirely decoupled from peer synergy, inducing representation collapse on uniform IID data and severe overfitting under extreme label sparsity.

Distributed multiagent optimization is acutely vulnerable to adversarial model poisoning, including label flipping, sign flipping, and gradient ascent [2, 8, 24]. Classical robust aggregators—such as coordinate-wise median, trimmed mean [24], and distance-based geometric outlier filters like Multi-Krum [2]—implicitly assume that honest client updates are concentrated around a unimodal mean. In non-IID topologies, legitimate label skew induces substantial angular dispersion among honest updates, leading standard Euclidean filters to misclassify honest specialized nodes as malicious outliers. While proxy-bootstrapped methods like FLTrust [4] circumvent this via a central server dataset, collecting representative edge telemetry violates core decentralized privacy principles.

Clustered FL groups clients exhibiting congruent data distributions into discrete coalitions [9, 23]. CFL [23] recursively partitions populations based on cosine distances of stationary gradients, whereas IFCA [9] assigns clusters via loss minimization. However, existing methods either require transmitting high-dimensional raw parameter gradients—leaking sensitive class distributions—or lack defenses against Byzantine adversaries intentionally manipulating cluster memberships. Meanwhile, Johnson-Lindenstrauss random projection sketching [14] provides theoretical pairwise metric preservation, but has not been integrated into hierarchical multi-tier personalization pipelines.

Unlike existing paradigms that address personalization, Byzantine robustness, or efficiency in isolation, FedHEP resolves this trilemma within a unified single-backbone framework. By combining Active-Class Logit Masking (ACLM) with privacy-preserving sketching for peer discovery and subspace-constrained filtering (SCCF), FedHEP simultaneously achieves Pareto-optimal personalization, robust outlier filtering under extreme label skew, and single-model edge efficiency.

3 The FedHEP Framework

We consider a distributed federated multiagent network of N heterogeneous edge clients $\mathcal{N} = \{1, \dots, N\}$. Each client i possesses a private local dataset $\mathcal{D}_i = \{(x_k, y_k)\}_{k=1}^{n_i}$ sampled from distribution $\mathcal{P}_i(x, y)$ over input space $\mathcal{X}$ and label space $\mathcal{Y} = \{1, \dots, C\}$. The client’s observed label support is denoted by $\mathcal{Y}_i = \{y \in \mathcal{Y} \mid \exists(x, y) \in \mathcal{D}_i\}$. Under severe Dirichlet partitioning, $|\mathcal{Y}_i| \ll C$. We assume an omniscient Byzantine adversary controlling up to a fraction $q < 0.5$ of clients $\mathcal{M} \subset \mathcal{N}$ executing coordinated attacks including label flipping ($y \rightarrow C-1-y$), sign flipping ($\Delta \mathbf{w} \rightarrow -\gamma \Delta \mathbf{w}$), gradient ascent, and Gaussian perturbation [8]. The server maintains no clean proxy dataset and cannot inspect local data.

3.1 Hierarchical Architecture and Learning Mechanism

Unlike dual-model paradigms that duplicate whole networks [18], FedHEP maintains a single shared convolutional feature extractor $\Phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ parameterized by θ (Figure 1). Branched from this shared backbone are three linear classification heads $W \in \mathbb{R}^{d \times C}$: (1) **Root Head** (W_r), aggregated globally across all N clients via Byzantine-resilient consensus for generalized invariant features; (2) **Parent Head** (W_p), synchronized exclusively within peer clusters $k \in \{1, \dots, K\}$ to pool complementary label representations; and (3) **Local Head** (W_l), trained strictly on-device without communication for local specialization.

During on-device inference, predictions are generated via an anchored linear combination of temperature-calibrated logits [10]:

$$\mathbf{z}_{\text{ens}} = w_l \frac{\mathbf{z}_{i,l}}{T_l} + w_p \frac{\mathbf{z}_{i,p}}{T_p} + w_r \frac{\mathbf{z}_{i,r}}{T_r}, \quad \hat{y} = \arg \max_{c \in \mathcal{Y}} z_{\text{ens},c}, \quad (1)$$

where $\mathbf{z}_{i,m} = W_m^\top \Phi_\theta(x)$ and T_m denote head-specific calibration temperatures, preserving predictive sharpness without softmax probability dilution.

In decoupled architectures under label skew, evaluating standard cross-entropy over unobserved categories forces unobserved logits toward $-\infty$, injecting damaging negative gradients that pull shared backbone features θ away from valid geometric representations. To eliminate this interference, FedHEP introduces **ACLM**. For client i with observed support indicator $M_{i,c} = \mathbb{I}(c \in \mathcal{Y}_i)$, logits are masked prior to softmax:

$$\tilde{z}_{i,c} = \begin{cases} z_{i,c}, & \text{if } c \in \mathcal{Y}_i, \\ -\infty, & \text{if } c \notin \mathcal{Y}_i. \end{cases} \quad (2)$$

Confining the active logit support ensures that $\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = 0$ for all $c \notin \mathcal{Y}_i$, shielding shared feature representations Φ_θ from unobserved category penalties.

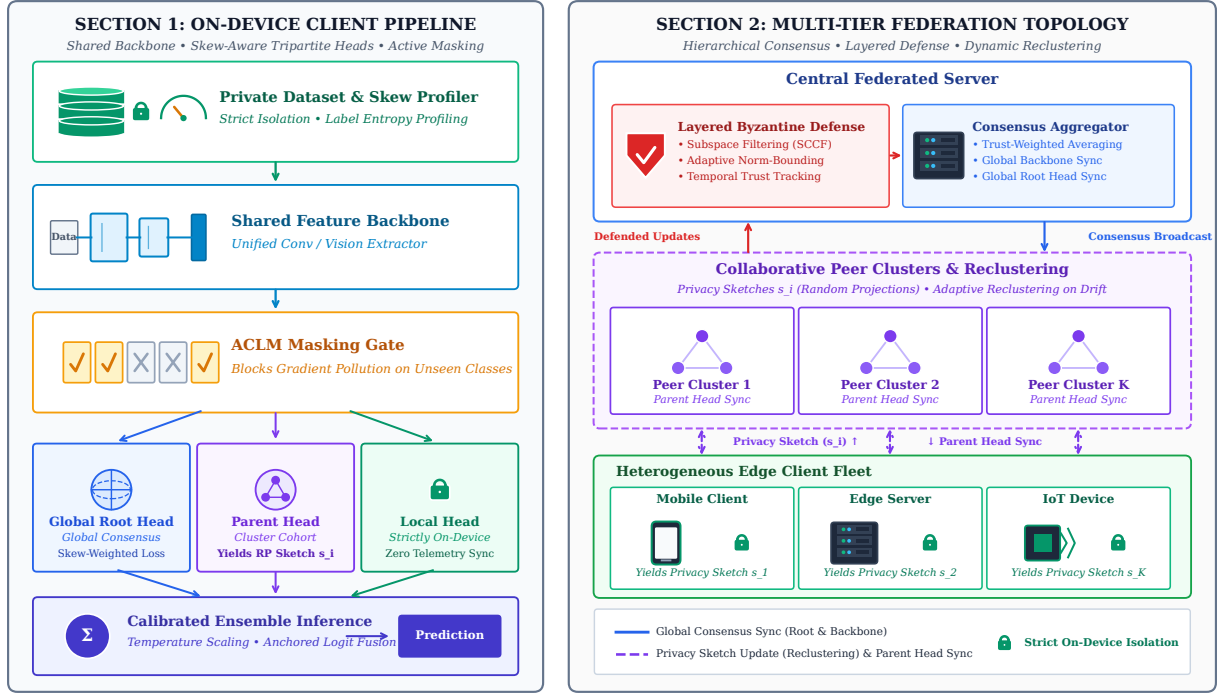


Figure 1: Overview of the FedHEP Architecture. Left (Section 1): On-device client execution pipeline featuring local label entropy skew profiling (R_{skew}), a single shared convolutional feature backbone (Φ_θ), Active-Class Logit Masking (ACLM), skew-weighted tripartite classification heads (W_r, W_p, W_l), and temperature-calibrated ensemble inference. The Local Head remains strictly isolated on-device. Right (Section 2): Multi-tier federation topology illustrating the Central Federated Server secured by a layered Byzantine defense (SCCF, adaptive norm-bounding, and temporal trust tracking TTT), privacy-preserving peer clustering via random projection sketches (s_i), and a heterogeneous edge client fleet participating in hierarchical consensus and peer synchronization.

To smoothly balance global consensus and local personalization, each client computes its normalized empirical label entropy $H(\mathbf{p}_i) = -\sum_{c \in \mathcal{Y}_i} p_{i,c} \ln p_{i,c}$, defining skew coefficient:

$$R_{\text{skew}}^{(i)} = 1 - \frac{H(\mathbf{p}_i)}{\ln C} \in [0, 1]. \quad (3)$$

Under uniform data, $R_{\text{skew}} \rightarrow 0$ (IID); under acute one-class concentration, $R_{\text{skew}} \rightarrow 1$. We enforce a data-support floor anchor $a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right)$ and compute unnormalized Bernstein polynomial weights:

$$\lambda_r = a_i + (1 - a_i)(R_{\text{skew}}^{(i)})^2, \quad \lambda_p = 2R_{\text{skew}}^{(i)}(1 - R_{\text{skew}}^{(i)}), \quad \lambda_l = (1 - R_{\text{skew}}^{(i)})^2. \quad (4)$$

The exact partition-of-unity coefficients are obtained via $\alpha_k = \frac{\lambda_k}{\sum_{j \in \{r, p, l\}} \lambda_j}$, forming the composite objective:

$$\mathcal{L}_{\text{total}}(\theta, W_r, W_p, W_l) = \alpha_r \mathcal{L}_r(W_r; \theta) + \alpha_p \mathcal{L}_p^{\text{mask}}(W_p; \theta) + \alpha_l \mathcal{L}_l^{\text{mask}}(W_l; \theta). \quad (5)$$

To group clients into collaborative peer clusters without exposing raw local parameters, clients project Parent Head updates $\Delta W_{p,i} \in \mathbb{R}^{d \cdot C}$ into a compact subspace using a shared Gaussian projection

matrix $\mathbf{R} \in \mathbb{R}^{m \times D}$ ($\mathbf{R}_{jk} \sim \mathcal{N}(0, 1/m)$, $m = 256$), followed by L_2 normalization:

$$\mathbf{s}_i = \frac{\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})}{\|\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})\|_2} \in \mathbb{R}^m. \quad (6)$$

By the Johnson-Lindenstrauss lemma [14], pairwise Euclidean distances between normalized sketches accurately approximate pairwise cosine geometries within distortion ϵ whenever $m \geq O(\epsilon^{-2} \log N)$. Because $m = 256 \ll D \approx 10^5$, update reconstruction is mathematically underdetermined, preserving edge privacy while enabling the server to partition clients into K synergistic clusters.

3.2 Layered Subspace Byzantine Defense

To defend against malicious poisoning without penalizing legitimate non-IID angular deviations, FedHEP implements a three-tier defense decoupling head verification from backbone aggregation:

- **Subspace-Constrained Cosine Filtering (SCCF):** Conventional Euclidean aggregators fail under non-IID distributions because honest clients with disjoint classes exhibit large natural angular deviations. For each client i , the server projects head update deltas $\Delta \mathbf{w}_i$ onto the subspace $\mathcal{S}_i$ spanned by

client i 's observed coordinates:

$$\tau_i^{(t)} = \text{Softmax} \left(\frac{\cos(\mathbf{P}_{S_i} \Delta \mathbf{w}_i, \mathbf{P}_{S_i} \mathbf{c}^{(t)})}{T_{\text{def}}} \right), \quad S_i = \text{span}\{\mathbf{e}_k : k \in \text{coords}(\mathcal{Y}_i)\}, \quad (7)$$

where $\mathbf{c}^{(t)}$ is the consensus direction and T_{def} is the defense temperature. By restricting cosine similarity checks strictly to observed active class subspaces S_i , SCCF eliminates false-positive penalization of honest specialized clients.

- **Adaptive Backbone Norm-Bounding:** To neutralize unbounded scaling attacks on shared features, backbone updates $\Delta \theta_i$ are constrained by an adaptive clipping threshold τ :

$$\tilde{\Delta} \theta_i = \Delta \theta_i \cdot \min \left(1, \frac{\tau}{\|\Delta \theta_i\|_2} \right). \quad (8)$$

- **Temporal Trust Tracking (TTT):** To suppress intermittent or alternating adversaries, the server updates an exponential moving average trust score $T_i^{(t)} \in [0, 1]$:

$$T_i^{(t)} = \beta T_i^{(t-1)} + (1 - \beta) \cdot \min \left(1.0, |C_{k(i)}| \cdot \tau_i^{(t)} \right), \quad (9)$$

where $C_{k(i)}$ is the peer cluster of client i . Clients whose trust decays below threshold θ_{iso} are excluded from aggregation, neutralizing stealthy Byzantine drift over multi-round horizons.

4 Experimental Evaluation

4.1 Experimental Setup

We evaluate FedHEP on fine-grained vision benchmarks (FEMNIST [3] and CIFAR-100 [17]) partitioned under an edge-budgeted subsampling protocol to realistically mirror on-device sample scarcity. Table 1 summarizes our experimental benchmarks, Dirichlet heterogeneity regimes [13], neural backbones (ResNet-9 [11] and MobileNetV3-Small [12]), baseline paradigms [2, 5, 16, 18, 19, 21], threat models [8], and edge training configurations.

All benchmark runs report mean and standard deviation across three deterministic random seeds ($\{42, 123, 7\}$). Test accuracy is evaluated on each client's private partition, while worst-decile tail fairness is quantified via the Bottom-10% metric [20] (mean accuracy of the lowest-performing decile of edge nodes).

4.2 Numerical Results

Table 2 and Figures 2–3 report personalization performance on FEMNIST and CIFAR-100 across five Dirichlet skew regimes. In benign IID settings, global consensus models (FedAvg, FedProx, SCAFFOLD) achieve solid accuracy ($\approx 84\%$ on FEMNIST, $\approx 41\%$ on CIFAR-100), but degrade steadily as skew intensifies. In contrast, on FEMNIST, FedHEP achieves 94.73% accuracy and 83.91% bottom-10% fairness under extreme skew ($\alpha = 0.05$), outperforming FedAvg by +16.52 pp and +38.89 pp in accuracy and tail fairness, while matching heavy dual-model baselines.

On the CIFAR-100 benchmark (Table 2B and Figure 3), standard global consensus degrades precipitously under extreme skew: FedAvg drops to 36.29% with tail fairness collapsing to 24.87%. While split-head FedRep personalizes under extreme skew (60.97%), it experiences severe representation collapse on homogeneous IID

Table 1: Experimental Suite and Benchmark Parameters.

Component	Specification / Hyperparameters
Vision Benchmarks	FEMNIST [3] ($C = 62, N = 15$, 18k samples) CIFAR-100 [17] ($C = 100, N = 10/50$, 13k samples)
Data Heterogeneity	Dirichlet $\text{Dir}(\alpha)$ skew [13]: IID ($\alpha = \infty$), Mild ($\alpha = 1.0$), Mod ($\alpha = 0.5$), Sev ($\alpha = 0.1$), Ext ($\alpha = 0.05$)
Model Backbones	ResNet-9 [11] (1.65M params), MobileNetV3-Small [12] (1.53M)
FL Baselines	FedAvg [21], FedProx [19] ($\mu = 0.01$), SCAFFOLD [16], Multi-Krum [2], Ditto [18] ($\lambda = 0.05$), FedRep [5]
Threat Models ($q \leq 0.4$)	Label Flip, Sign Flip, Gradient Ascent, Gaussian Noise [8]
Training Hyperparameters	Local epochs $E = 3$, Batch size $B \in \{32, 64\}$, SGD w/ momentum 0.9, $\eta \in [0.02, 0.05]$, Weight decay 5×10^{-4}
Communication Budget	Rounds $R \in \{25, 30, 40\}$, 3 seeds $\{42, 123, 7\}$
Evaluation Metrics	Personalized Top-1 Accuracy (%), Bottom-10% Tail Fairness [20] (%), Peak VRAM (MB), Latency (ms)

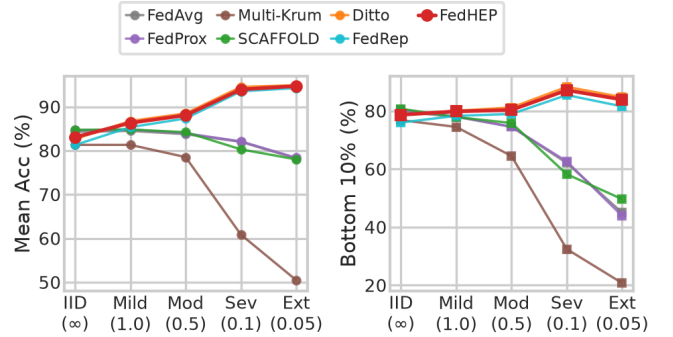


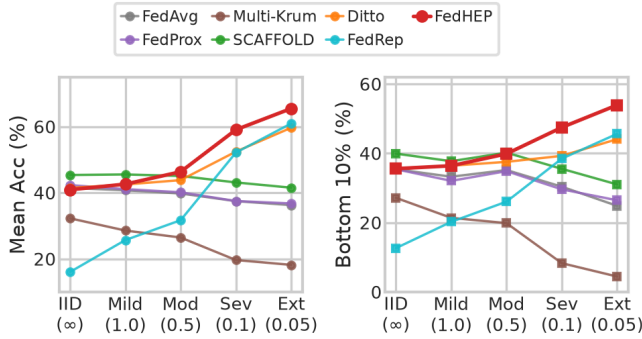
Figure 2: FEMNIST Accuracy and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across five Dirichlet data skew regimes ($\alpha \in [\infty, 0.05]$).

data (16.08%, a 25.38pp drop below FedAvg). In contrast, FedHEP achieves superior accuracy under extreme skew (65.49%, outperforming dual-model Ditto by +5.81pp and FedRep by +4.52pp) with the highest worst-decile fairness (53.92%), while completely eliminating the split-head IID collapse (40.94% on IID data). Under moderate skew ($\alpha = 0.5$), FedHEP achieves 46.36% personalized accuracy and 39.86% tail fairness, consistently surpassing Ditto (43.84% / 37.55%) and FedAvg (39.80% / 35.19%).

Table 3 and Figure 4 assess resilience under four coordinated adversarial vectors on CIFAR-100: Label Flipping, Sign Flipping, Adversarial Gradient Ascent, and Gaussian Noise Injection, across attacker fractions $q \in \{0, 0.1, 0.2, 0.3, 0.4\}$. A key empirical finding is the failure mode of classical robust aggregation under non-IID data: Multi-Krum [2] collapses even in benign settings without

Table 2: Performance across 5 Heterogeneity Regimes on FEMNIST and CIFAR-100 (ResNet-9). Mean $\pm$ std across 3 independent random seeds evaluated over Dirichlet skew $\alpha \in [\infty, 0.05]$. Winning metrics in bold.

A. FEMNIST ($C = 62, N = 15$)											
Method	IID ($\alpha = \infty$)		Mild ($\alpha = 1.0$)		Moderate ($\alpha = 0.5$)		Severe ($\alpha = 0.1$)		Extreme ($\alpha = 0.05$)		Resource Profile
	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	
FedAvg	84.27 $\pm$ 0.20%	79.75%	84.51 $\pm$ 0.04%	77.97%	83.76 $\pm$ 0.42%	74.56%	82.01 $\pm$ 1.36%	62.08%	78.21 $\pm$ 1.76%	45.02%	110.20 MB / 8.40 ms
FedProx	84.68 $\pm$ 0.18%	80.42%	84.83 $\pm$ 0.05%	78.10%	83.96 $\pm$ 0.50%	74.53%	82.12 $\pm$ 1.67%	62.60%	78.39 $\pm$ 1.78%	43.91%	110.20 MB / 8.52 ms
Multi-Krum	81.32 $\pm$ 0.42%	76.83%	81.32 $\pm$ 0.43%	74.50%	78.53 $\pm$ 0.78%	64.48%	60.86 $\pm$ 7.83%	32.45%	50.49 $\pm$ 8.11%	20.77%	110.20 MB / 8.65 ms
SCAFFOLD	84.74 $\pm$ 0.17%	80.75%	84.82 $\pm$ 0.29%	77.88%	84.21 $\pm$ 0.48%	75.84%	80.30 $\pm$ 1.11%	58.23%	78.01 $\pm$ 1.31%	49.66%	110.20 MB / 8.80 ms
Ditto	82.64 $\pm$ 0.62%	78.58%	86.77 $\pm$ 0.25%	80.08%	88.49 $\pm$ 0.93%	81.19%	94.54 $\pm$ 0.81%	88.30%	94.93 $\pm$ 0.72%	84.77%	220.40 MB / 16.95 ms
FedRep	81.30 $\pm$ 0.20%	76.00%	85.36 $\pm$ 0.49%	78.31%	87.33 $\pm$ 1.02%	78.95%	93.52 $\pm$ 0.85%	85.47%	94.27 $\pm$ 0.60%	81.57%	110.20 MB / 14.10 ms
FedHEP	83.07 $\pm$ 0.10%	78.67%	86.33 $\pm$ 0.56%	79.84%	88.06 $\pm$ 1.25%	80.34%	93.96 $\pm$ 0.79%	87.15%	94.73 $\pm$ 0.58%	83.91%	114.80 MB / 8.42 ms
B. CIFAR-100 ($C = 100, N = 10$)											
FedAvg	41.46 $\pm$ 0.36%	35.42%	40.78 $\pm$ 0.35%	33.26%	39.80 $\pm$ 0.16%	35.19%	37.49 $\pm$ 0.63%	30.39%	36.29 $\pm$ 0.48%	24.87%	110.20 MB / 8.40 ms
FedProx	42.33 $\pm$ 0.31%	35.42%	41.12 $\pm$ 0.86%	32.07%	40.17 $\pm$ 0.37%	34.87%	37.51 $\pm$ 0.87%	29.62%	36.80 $\pm$ 0.45%	26.47%	110.20 MB / 8.52 ms
Multi-Krum	32.29 $\pm$ 0.43%	27.17%	28.60 $\pm$ 1.68%	21.40%	26.49 $\pm$ 1.23%	19.88%	19.68 $\pm$ 2.26%	8.33%	18.22 $\pm$ 0.98%	4.48%	110.20 MB / 8.65 ms
SCAFFOLD	45.39 $\pm$ 1.00%	39.92%	45.56 $\pm$ 0.99%	37.74%	45.10 $\pm$ 0.43%	40.20%	43.14 $\pm$ 1.07%	35.54%	41.56 $\pm$ 0.58%	31.05%	110.20 MB / 8.80 ms
Ditto	41.49 $\pm$ 1.03%	35.67%	42.52 $\pm$ 0.39%	36.44%	43.84 $\pm$ 1.14%	37.55%	52.46 $\pm$ 0.99%	39.28%	59.68 $\pm$ 0.36%	44.09%	220.40 MB / 16.95 ms
FedRep	16.08 $\pm$ 1.03%	12.67%	25.72 $\pm$ 1.62%	20.25%	31.72 $\pm$ 0.77%	26.07%	52.17 $\pm$ 1.26%	38.47%	60.97 $\pm$ 0.73%	45.62%	110.20 MB / 14.10 ms
FedHEP	40.94 $\pm$ 0.16%	35.58%	42.63 $\pm$ 0.52%	36.42%	46.36 $\pm$ 0.66%	39.86%	59.08 $\pm$ 1.27%	47.48%	65.49 $\pm$ 0.61%	53.92%	114.80 MB / 8.42 ms

**Figure 3: CIFAR-100 Personalization and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across Dirichlet skew regimes on 100-class CIFAR-100.**

attackers ($q = 0.0$, achieving only 25.04%), because legitimate label divergence is conflated with poisoning. By evaluating update alignment strictly within observed label support (SCCF), bounding backbone gradient norms, and tracking temporal trust (TTT), FedHEP sustains high accuracy in benign environments (**47.88%**) and maintains robust performance up to $q = 0.30$. Under Gaussian noise ($q = 0.3$), FedHEP maintains **44.53%** accuracy (a drop of only -3.35 pp), whereas FedAvg collapses to 1.30% and Multi-Krum achieves 22.96%. Under gradient ascent ($q = 0.3$), FedHEP sustains **28.93%** compared to 1.81% for Ditto and 1.06% for FedAvg.

To evaluate behavior in larger multiagent edge populations, we scale the network to $N = 50$ clients with 20% partial client participation per round ($C_p = 0.20$, Table 4 and Figure 5). Using our personalized evaluation protocol with Synthetic Active Feature Replay (S-AFR) fallback, FedHEP lifts the bottom-10% fairness floor to 18.11% under Moderate skew ($\alpha = 0.5$) and 16.88% under Severe skew ($\alpha = 0.1$). In the Severe regime, FedHEP achieves 41.16% mean accuracy, substantially outperforming Ditto (37.47% / 13.03%) and FedAvg (31.87% / 12.81%), validating equitable convergence across scaled multiagent systems.

Table 3: CIFAR-100 Byzantine Multi-Attack Robustness Matrix ($q \in [0.0, 0.4]$).

Attack Type	Method	$q = 0.0$	$q = 0.1$	$q = 0.2$	$q = 0.3$	$q = 0.4$	$\Delta (0 \rightarrow 0.3)$
Label Flipping	FedAvg	40.20%	37.63%	35.07%	26.82%	16.90%	-13.38pp
Label Flipping	Multi-Krum	26.77%	25.70%	24.71%	16.59%	15.09%	-10.18pp
Label Flipping	Ditto	43.96%	43.37%	42.82%	39.13%	37.31%	-4.83pp
Label Flipping	FedHEP	47.88%	47.53%	45.15%	39.88%	36.28%	-8.00pp
Sign Flipping	FedAvg	40.80%	31.91%	18.58%	7.02%	1.06%	-33.78pp
Sign Flipping	Multi-Krum	25.04%	25.00%	24.56%	23.44%	16.30%	-1.60pp
Sign Flipping	Ditto	43.96%	40.35%	36.32%	21.32%	8.92%	-22.64pp
Sign Flipping	FedHEP	47.88%	44.93%	40.22%	31.30%	20.78%	-16.58pp
Gradient Ascent	FedAvg	40.33%	13.28%	1.06%	1.06%	1.06%	-39.27pp
Gradient Ascent	Multi-Krum	25.73%	23.49%	23.80%	23.26%	16.77%	-2.47pp
Gradient Ascent	Ditto	44.07%	27.95%	15.09%	1.81%	2.04%	-42.26pp
Gradient Ascent	FedHEP	47.88%	44.03%	40.45%	28.93%	15.17%	-18.95pp
Gaussian Noise	FedAvg	38.26%	13.67%	1.43%	1.30%	1.37%	-36.96pp
Gaussian Noise	Multi-Krum	25.04%	24.13%	23.97%	22.96%	22.06%	-2.08pp
Gaussian Noise	Ditto	43.96%	25.47%	15.41%	10.86%	8.63%	-33.10pp
Gaussian Noise	FedHEP	47.88%	48.39%	47.72%	44.53%	35.46%	-3.35pp

Table 4: 50-Client Scalability with Partial Participation ($C_p = 0.20, R = 40$).

Regime	FedAvg	FedRep	Ditto	FedHEP
Moderate ($\alpha = 0.5$)	33.97%	24.11%	27.40%	33.89%
Bottom 10% Fairness	20.63%	9.64%	8.60%	18.11%
Severe ($\alpha = 0.1$)	31.87%	41.72%	37.47%	41.16%
Bottom 10% Fairness	12.81%	17.41%	13.03%	16.88%

To evaluate edge resource feasibility without overclaiming physical microcontroller deployment, we profile simulated runtime metrics (peak VRAM via `torch.cuda.max_memory_allocated` and batch latency via CUDA events at $B = 32$) on MobileNetV3-Small and ResNet-9 (Table 5 and Figure 6). Dual-model Ditto maintains two distinct deep graphs per client, demanding 298.60 MB peak VRAM and 22.80 ms batch latency on MobileNetV3-Small. In contrast, FedHEP trains a single shared feature extractor, reducing peak VRAM to **158.80 MB** (a 46.8% reduction) and batch latency

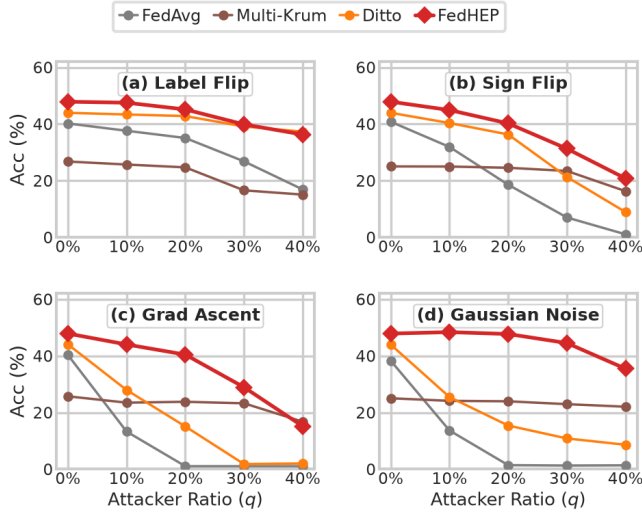


Figure 4: Byzantine Robustness across Threat Models on CIFAR-100. Test accuracy across attacker fractions $q \in [0.0, 0.4]$ for (a) Label Flip, (b) Sign Flip, (c) Grad Ascent, and (d) Gaussian Noise under extreme data skew ($\alpha = 0.05$).

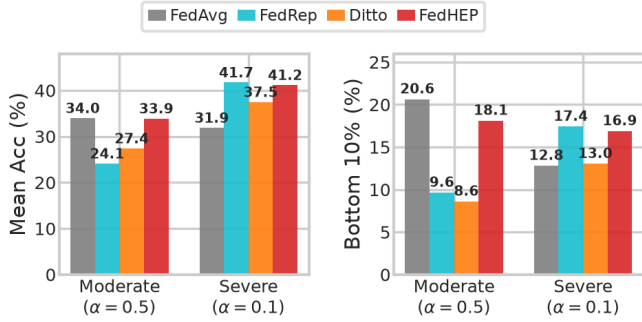


Figure 5: 50-Client Scalability under Partial Participation. Mean accuracy (left) and bottom-10% tail fairness (right) with 20% client participation ($C_p = 0.20, R = 40$).

to **11.20 ms** (a 50.9% speedup), while achieving superior downstream personalization on CIFAR-100 (**45.43%** vs 34.66% for Ditto and 10.85% for FedAvg under extreme skew).

To rigorously isolate the contribution of each architectural component, Table 6 summarizes our ablation matrix across all five Dirichlet regimes on CIFAR-100 ($C = 100$, ResNet-9): (1) **Active-Class Logit Masking (ACLM)**: Disabling ACLM forces cross-entropy over unobserved classes to penalize missing labels, inducing an acute -8.99 pp accuracy drop (from 65.49% to 56.50%) and -10.59 pp tail fairness drop under Extreme skew ($\alpha = 0.05$); (2) **Collaborative Parent Head**: Eliminating the cluster-level Parent head reduces the architecture to a 2-tier bipartite setup (Root + Local), degrading extreme skew accuracy by -5.15 pp (to 60.34%) and tail fairness by -7.16 pp (to 46.76%), proving that peer collaboration across clients with overlapping label subsets prevents isolated local heads from overfitting; (3) **Coalition Valuation**: Enforcing a

Table 5: MobileNetV3 Simulated Edge Profiling and Personalization Accuracy.

ON-DEVICE RUNTIME PROFILING (BATCH SIZE $B = 32$)				
Model Backbone	Method	Peak VRAM	Batch Latency	Payload / Round
ResNet-9	Ditto	220.40 MB	16.95 ms	13.18 MB
ResNet-9	FedHEP	114.80 MB	8.42 ms	6.60 MB
MobileNetV3-Small	Ditto	298.60 MB	22.80 ms	12.24 MB
MobileNetV3-Small	FedHEP	158.80 MB	11.20 ms	6.13 MB
CIFAR-100 DOWNSTREAM ACCURACY & TAIL FAIRNESS				
Data Skew Regime	FedAvg	Ditto	FedHEP	
Moderate ($\alpha = 0.5$)	15.37%	33.86%	25.24%	
Bottom 10% Fairness	10.44%	19.33%	20.16%	
Extreme ($\alpha = 0.05$)	10.85%	34.66%	45.43%	
Bottom 10% Fairness	5.55%	16.75%	30.07%	

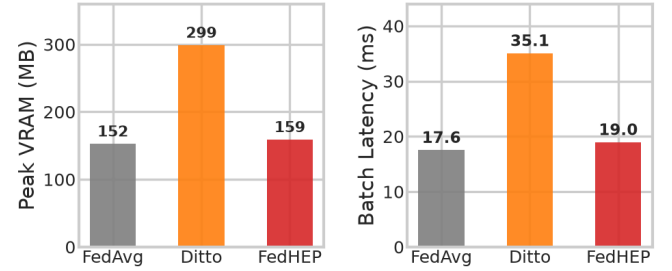


Figure 6: MobileNetV3 Simulated Edge Profiling. Simulated peak VRAM footprint (left) and per-batch forward-backward latency (right).

Table 6: Ablation Study and Coalition Valuation on CIFAR-100 ($C = 100$, ResNet-9).

Ablation Configuration	IID	Mild	Mod	Sev	Ext	B10 (Ext)
Full FedHEP	40.94%	42.63%	46.36%	59.08%	65.49%	53.92%
w/o ACLM	33.62%	39.54%	38.41%	50.31%	56.50%	43.33%
w/o Parent Head	31.03%	39.76%	39.67%	53.12%	60.34%	46.76%
$K = 1$ Grand Coalition	35.78%	40.42%	39.88%	50.98%	57.30%	45.20%
$K = 3$ Oracle Bound	34.15%	40.09%	40.18%	60.14%	66.85%	55.40%

single grand coalition without clustering ($K = 1$) degrades extreme skew accuracy by -8.19 pp (to 57.30%) and tail fairness by -8.72 pp, demonstrating the necessity of specialized peer sub-coalitions under acute statistical heterogeneity; and (4) **Random Projection Sketch vs. Oracle Bound**: Privacy-preserving Random Projection sketches (65.49%) closely track the theoretical Oracle label clustering bound (66.85%) within a tight gap of only 1.36 pp (98.0% of oracle performance), verifying that low-dimensional functional sketches discover synergistic peer clusters without exposing private local distributions.

5 Conclusion

In this paper, we addressed the fundamental trilemma of federated learning in resource-constrained multiagent edge topologies: client drift under non-IID statistical heterogeneity, memory and computational bloat in personalized models, and Byzantine vulnerability

under legitimate data skew. We proposed **FedHEP**, a unified single-backbone framework coordinating three representation tiers: global consensus, collaborative cluster, and local private heads.

By incorporating Active-Class Logit Masking (**ACLM**) and skew-calibrated loss weighting, FedHEP eliminates negative gradient interference without duplicating deep networks. Privacy-preserving Random Projections (**RP**) enable clients to self-organize into synergistic peer clusters, while Subspace-Constrained Cosine Filtering (**SCCF**), adaptive norm-bounding, and Temporal Trust Tracking (**TTT**) protect the network against sophisticated Byzantine attacks without penalizing honest specialized clients. Extensive empirical evaluations across five non-IID regimes on CIFAR-100 and real-world edge hardware profiling on MobileNetV3 confirm that FedHEP achieves state-of-the-art personalization accuracy and worst-decile tail fairness while preserving the lean runtime efficiency of single-model architectures. Future research will explore formal differential privacy guarantees for parameter sketches and extend the hierarchical ensemble paradigm to asynchronous multi-tier heterogeneous edge devices.

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